Suggested problems in Hughes-Hallett, §12.1: 1-37 odd

Problem 1: The temperature adjusted for wind chill is a temperature which tells you how cold it feels, as a result of the combination of wind and temperature; see the table below.

Wind Speed (mph) \ Temperature (°F)	35	30	25	20	15	10	5	0
5	31	25	19	13	7	1	−5	−11
10	27	21	15	9	3	−4	−10	−16
15	25	19	13	6	0	−7	−13	−19
20	24	17	11	4	−2	−9	−15	−22
25	23	16	9	3	−4	−11	−17	−24

(a) Express the dependency of wind chill on wind and temperature in function notation.

$C = f(w, t)$, where C is the wind chill, w is the wind speed, and t is the temperature.

(b) If the temperature is 0°F and the wind speed is 15 mph, how cold does it feel?

−19°F

(c) If the temperature is 35°F, what wind speed makes it feel like 24°F?

20 mph

Problem 2: A car rental company charges \$40 per day and 15 cents a mile for its cars.

(a) Write a formula for the cost, C, of renting a car as a function, h, of the number of days, d, and the number of miles driven, m.

$C = h(d, m) = 40d + 0.15m$

(b) Find $h(5, 300)$ and interpret it.

We have

$$h(5, 300) = 40 \cdot 5 + 0.15 \cdot 300 = 245$$

This means that it cost \$245 to rent a car for 5 days, during which 300 miles are driven.

Problem 3: Suppose $f(x, y) = x^2y + \sqrt{x}$.

(a) $f(1, 2)$ 3

(b) $f(9, 4)$ 327

Problem 4: Consider the three points

$$A = (1.3, -2.7, 0),$$
$$B = (0.9, 0, 3.2),$$
$$C = (2.5, 0.1, -0.3),$$

in $\mathbb{R}^3$.

(a) Which one is closest to the yz-plane?

B

(b) Which one lies on the xy-plane?

A

(c) Which one is farthest from the xz-plane?

A